

Software Engineering

SEN 220

Project Report



Group members:

Maarij Niazi (01-134242-059)

Laiba Zainab (01-134242-058)

Duaa Rasheed (01-134242-040)

Ayesha Safeer (01-134242-031)

Submitted to: Ma'am Aima Zahoor

**BAHRIA UNIVERSITY ISLAMABAD
DEPARTMENT OF COMPUTER SCIENCE**

Digital Lost and Found System (Campus)

Problem Statement:

Students at university campuses are often in the habit of losing their personal items like identity cards, books, accessories, etc. At present, there is no mechanism available for reporting or claiming lost property on campus, which is inefficient and results in losses of valuable assets.

This project attempts to develop a system which allows users to report their lost items, upload found items and make matches efficiently.

Project Scope:

System features include:

- User reporting of lost items.
- User submission of found items.
- Automatic or manual matching of lost and found items.
- Notification of users on possible matches.
- Maintenance of records of all reports.
- Admin moderation.

Functional Requirements:

1. Users Signup & Access:

Students can register with their student ID or email address. Users can sign in and sign out.

2. Report Missing Item:

The user can submit details such as item name, description, location, date and possible image.

3. Report Lost Item:

The user can upload information about items found.

4. Things to look for:

People can search for lost or found items by category or keywords.

5. Browse item listings:

Users can view all lost and found items.

6. Speak to the User:

The person who uploaded the item is available for contact.

7. Administrative services:

Admin can remove inappropriate or fraudulent posts. Users can be managed by Admin.

8. Edit or Delete Post:

Users can edit and delete their own posts.

Non-Functional Requirements:

1. Usability:

The system should have a user-friendly interface.

2. Performance:

The system must respond in two to three seconds.

3. Security:

User information needs to be protected. Items may only be posted by users that are registered.

4. Availability:

The system should be available 24/7.

5. Scalability:

Must be able to support growing number of users and posts.

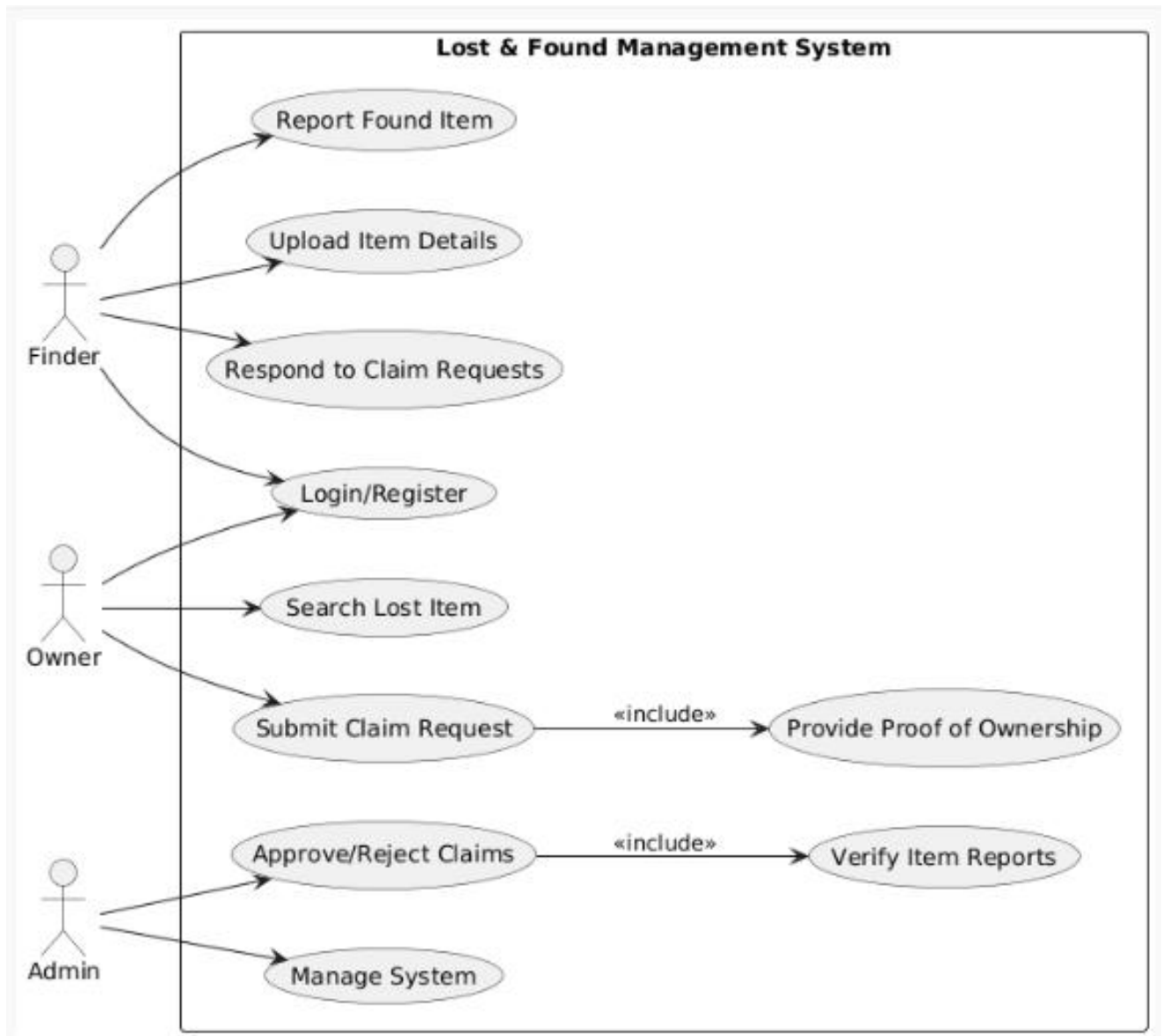
6. Reliability:

Data should not be lost or modified.

7. Compatibility:

Desktop and mobile browser should be compatible

Use Case Diagram:



Use case description tables:

1. Use Case: Login/Register

Field	Details
Use Case Name	Register/Login User
Actor	User
Description	User creates a new account or logs into the system using credentials.
Trigger	User selects “Register” or “Login”
Normal Flow	• User opens registration/login page • If new user, selects register and enters details (name, email, password) • System validates data and creates account • If existing user, enters email and password • System verifies credentials • User is logged in successfully
Alternative Flow	• Missing or invalid data shows error message • Email already exists shows duplication error during registration • Weak password is rejected by system • Invalid login credentials show error message • Unregistered account shows error message

2. Use Case: Report Found Item

Field	Details
Use Case Name	Report Found Item
Actor	Finder
Description	User submits details of an item found on campus.
Trigger	User selects “Report Found Item”
Normal Flow	• User logs in • Opens form • Enters item details • Submits report • System saves item
Alternative Flow	• Invalid input shows an error message • Missing data causes the form to be rejected.

3. Use Case: Search lost Item

Field	Details
Use Case Name	Search lost item
Actor	Owner
Description	User searches items using keywords or category.
Trigger	User enters search query
Normal Flow	• User opens search page • Enters keyword or category

Field	Details
	• System searches database • Results are displayed
Alternative Flow	• No results found, system shows “No match found” • Empty search prompts user to enter input

4. Use Case: Manage System

Field	Details
Use Case Name	Manage System
Actor	Admin
Description	Admin manages users and posts (delete, monitor, approve).
Trigger	Admin logs into system
Normal Flow	• Admin logs in • Opens dashboard • Manages users and posts • Removes or updates records
Alternative Flow	• Unauthorized access is denied by the system • Invalid action is blocked by the system

5. Use case: Upload item details

Field	Details
Use Case Name	Upload item details
Actor	Finder
Description	User reports a missing item with details like name, description, location, date, and image.
Trigger	User selects “Report Lost Item”
Normal Flow	• User logs in • Opens report form • Enters item details • Submits report • 5. System stores data
Alternative Flow	• Missing fields trigger an error message • Invalid image is rejected by the system • If not logged in, access is denied

6. Use case: Submit claim requests

Field	Details
Use Case Name	Submit Claim Requests
Actor	Owner
Description	User submits a request to claim ownership of a found item.
Trigger	User clicks “Submit Claim” option
Normal Flow	• User logs in • Opens item details • Clicks claim request option • Enters claim message/details • Submits request

Field	Details
	System sends request to finder/admin
Alternative Flow	- Not logged in results in access denied - Empty claim details show error message - Claim submission failure shows retry message

7. Use case: Respond to claim requests

Field	Details
Use Case Name	Respond to Claim Requests
Actor	User
Description	Finder or admin responds to a claim request submitted by another user.
Trigger	User opens received claim request
Normal Flow	- User logs in - Opens claim requests section - Selects a request - Types response message - Sends response - System delivers response to claimant
Alternative Flow	- Unauthorized access is denied - Request not found shows error message - System error prevents message delivery

8. Use case: Approve/Reject claims

Field	Details
Use Case Name	Approve/Reject Claims
Actor	Admin
Description	Admin reviews submitted claims and approves or rejects them based on verification.

Field	Details
Trigger	Admin selects a claim request
Normal Flow	• Admin logs in • Opens claims dashboard • Reviews claim details and proof • Chooses to approve or reject option • System updates claim status • • User receives notification
Alternative Flow	• Unauthorized access is denied • Claim not found shows error message • System failure prevents status update

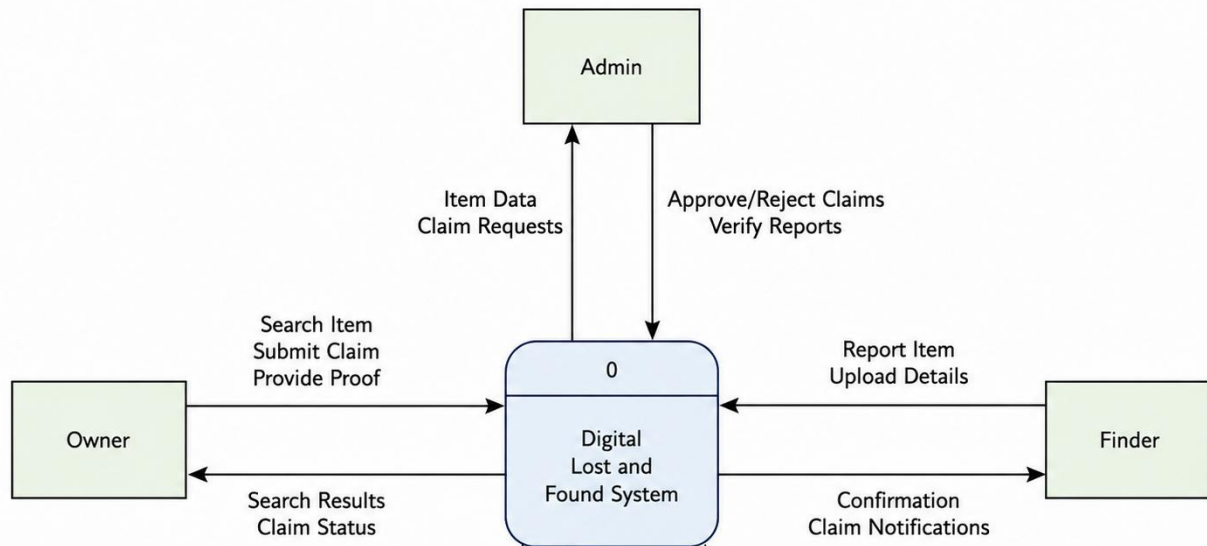
9. Use case: Provide proof of ownership

Field	Details
Use Case Name	Provide Proof of Ownership
Actor	Owner
Description	User provides evidence to prove ownership of a lost item.
Trigger	User selects “Provide Proof” option
Normal Flow	• User logs in • Opens claim request • Uploads proof documents/images • Adds additional details • Submits proof • System stores proof for verification
Alternative Flow	• Invalid file format is rejected • Missing proof details show error message • Upload failure shows retry option

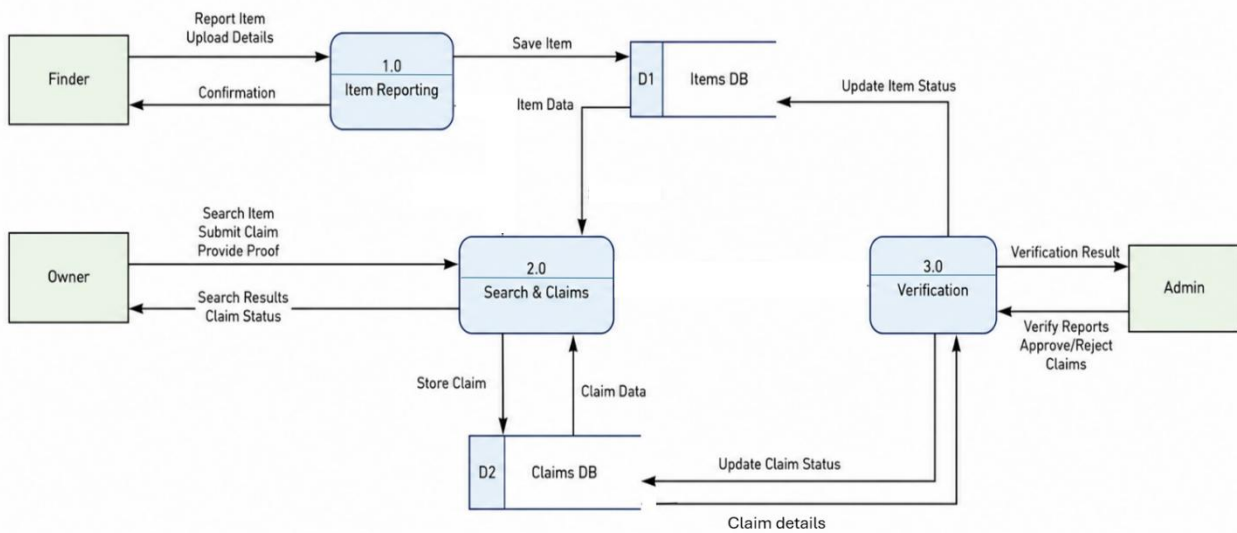
10. Use case: Verify item reports

Field	Details
Use Case Name	Verify Item Reports
Actor	Admin
Description	Admin verifies whether lost or found item reports are valid and appropriate.
Trigger	Admin opens verification section
Normal Flow	• Admin logs in • Opens item reports list • Reviews item details and uploaded data • Verifies report authenticity • Approves or flags report • System updates report status
Alternative Flow	• Unauthorized access is denied • Report not found shows error message • System error prevents verification process

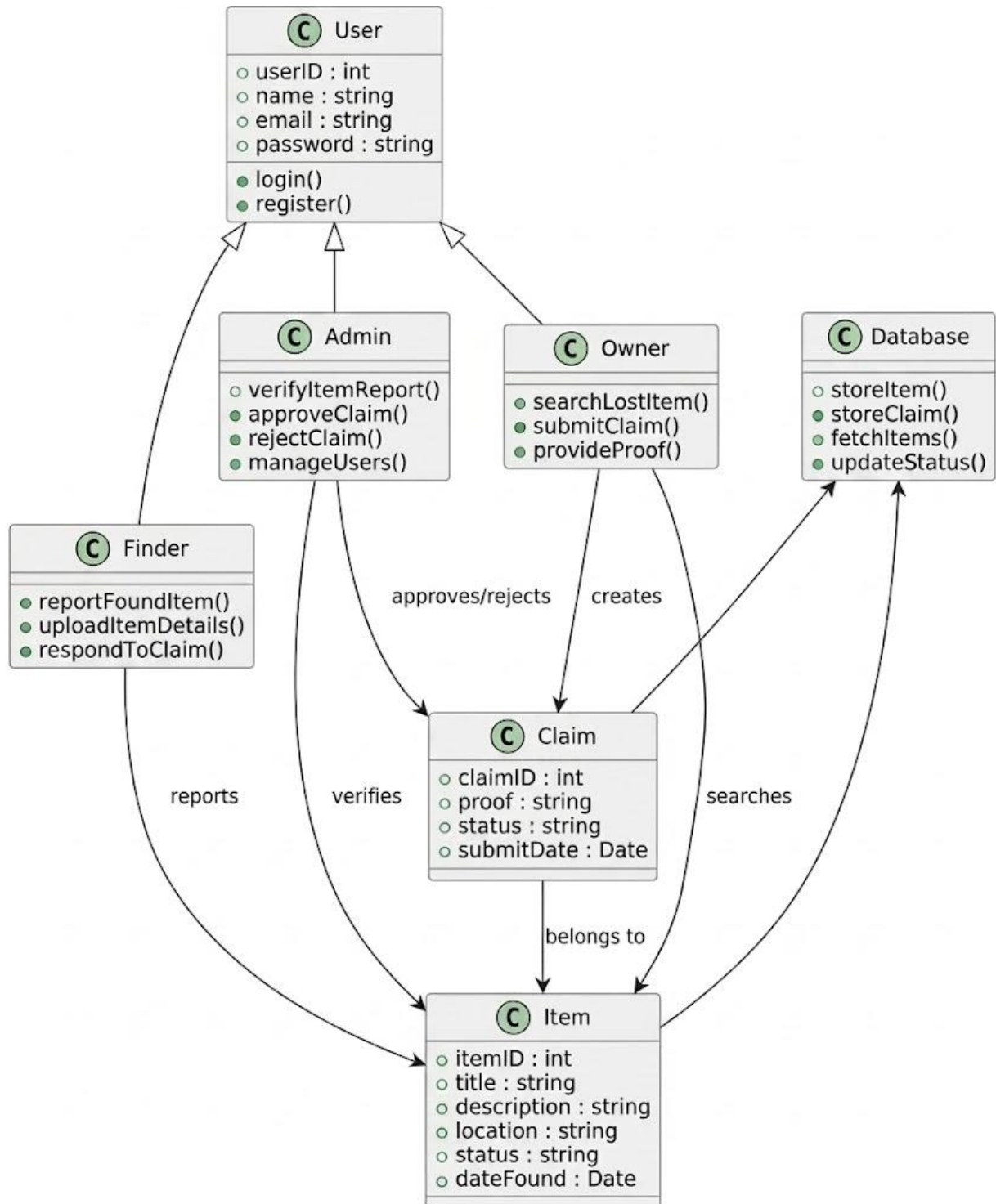
Context diagram:



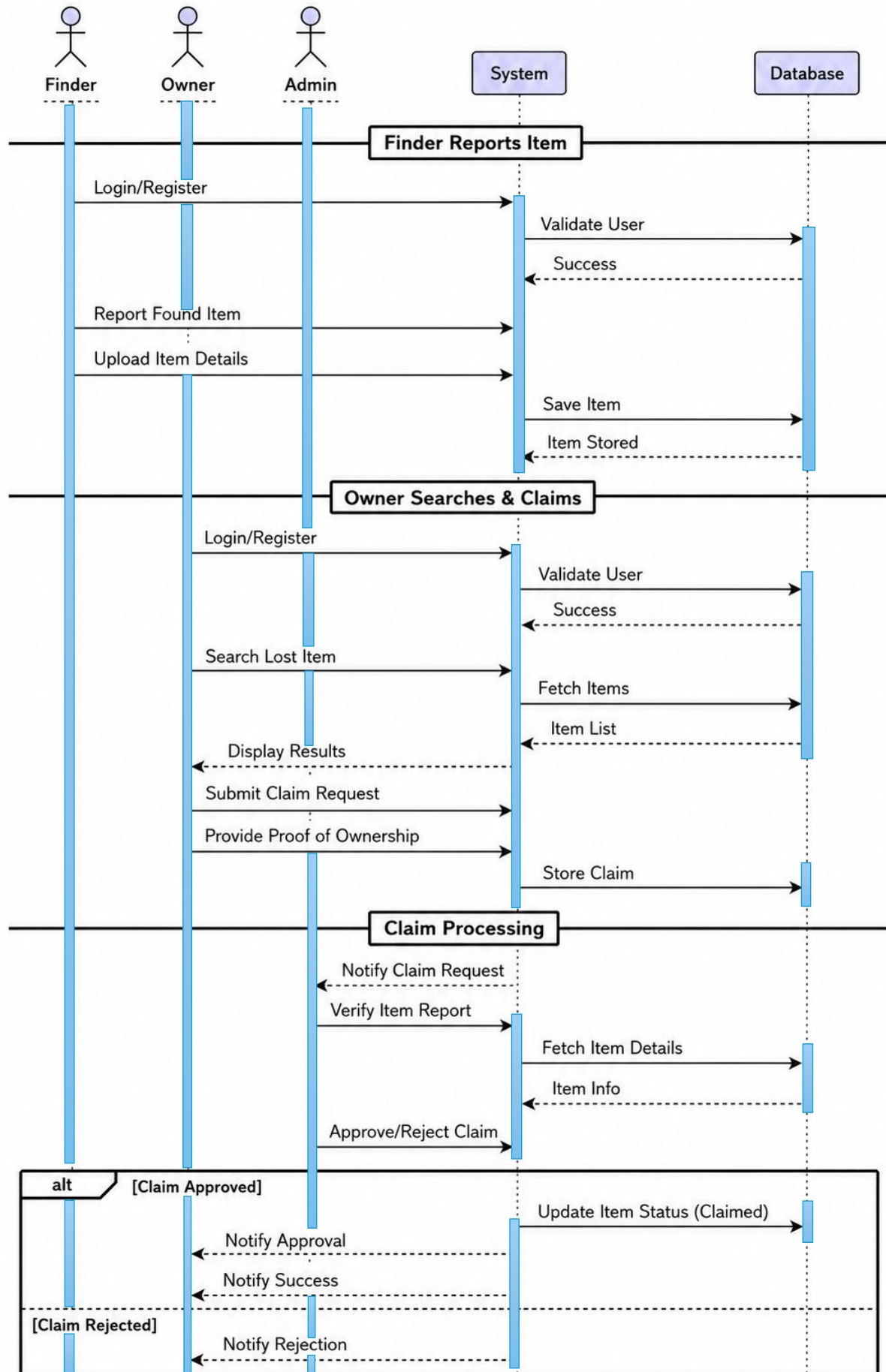
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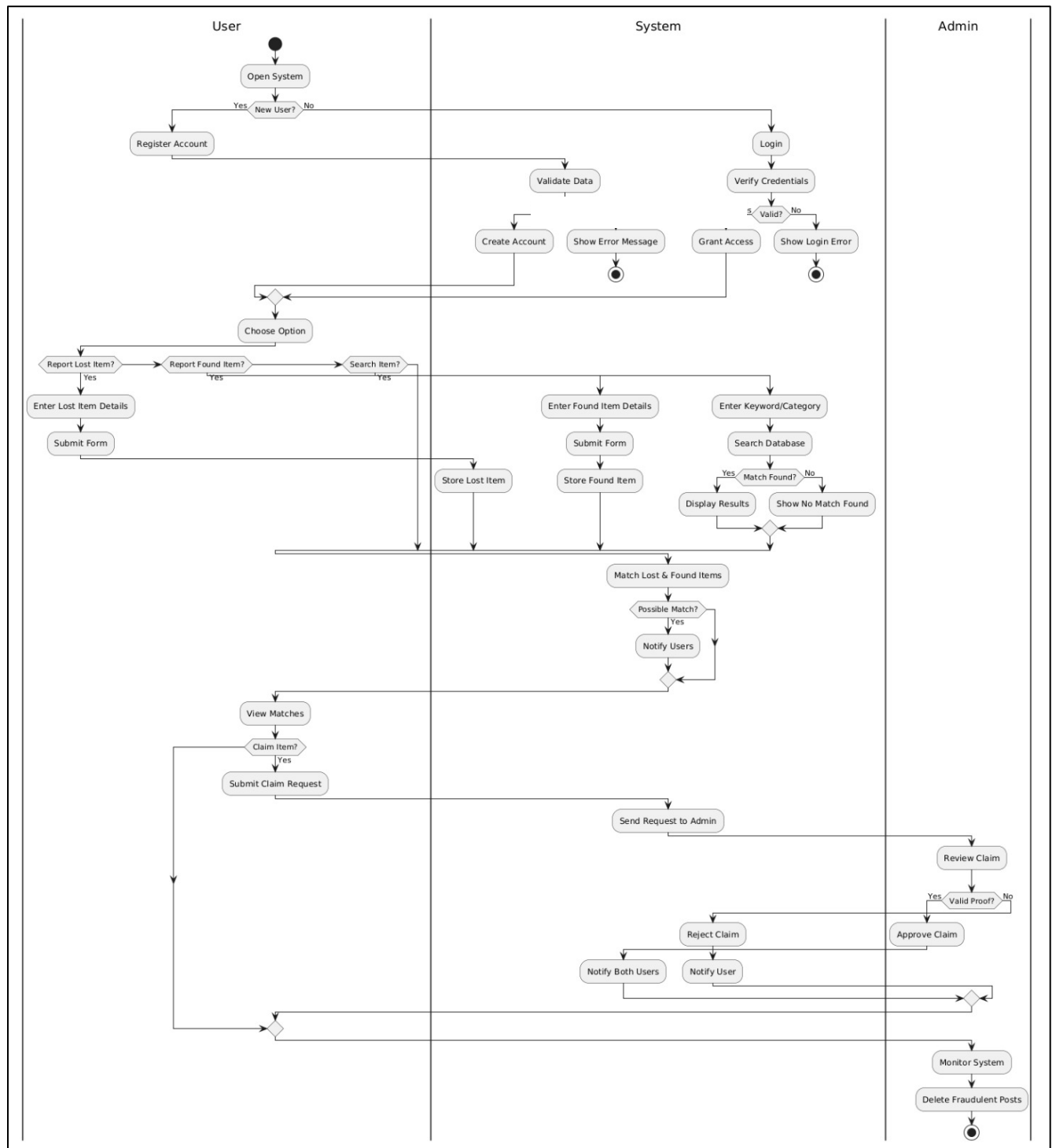
Class diagram:



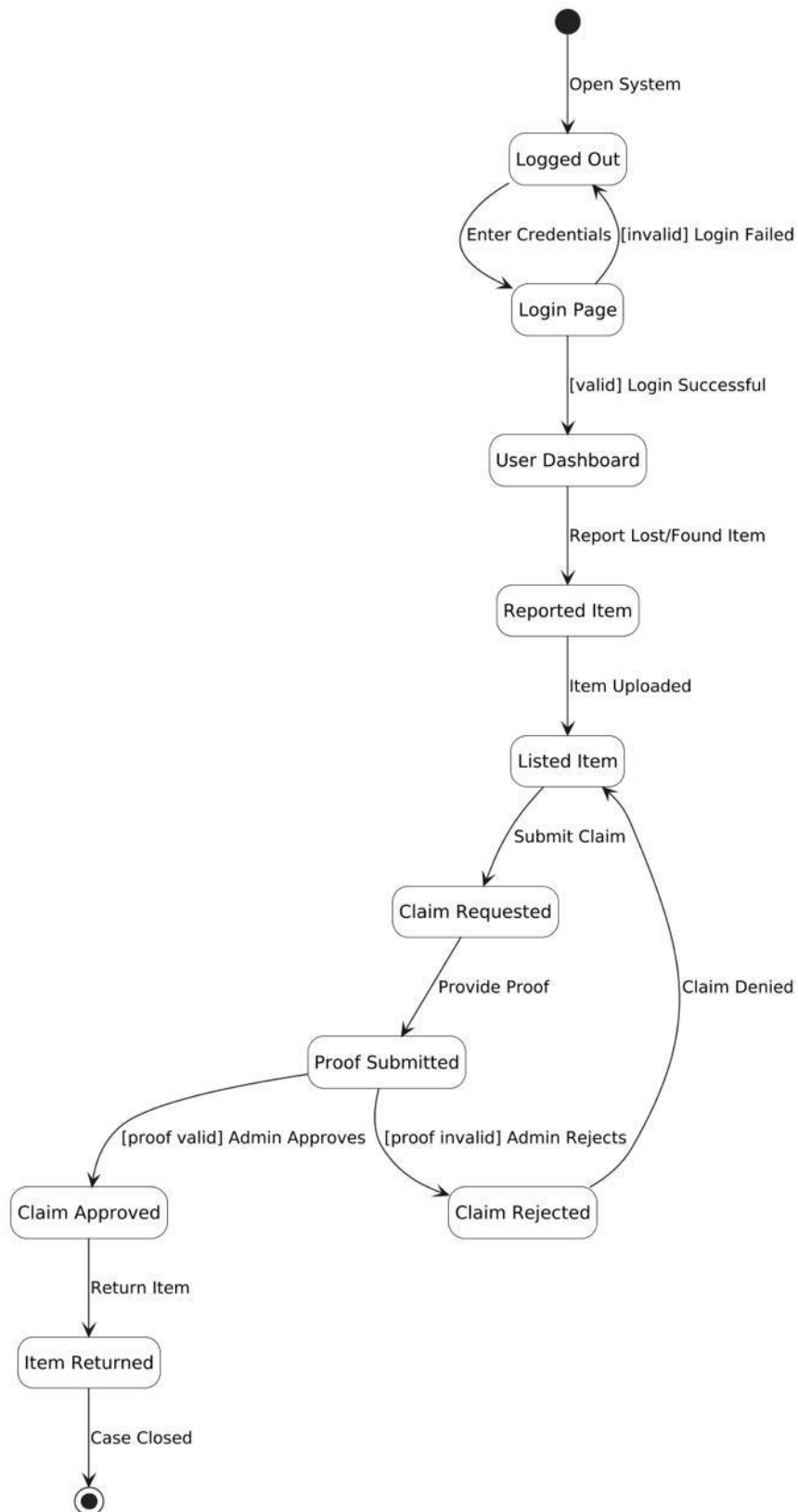
Sequence diagram:



Activity diagram:



State diagram:



UML Diagrams refined:

Following the completion of the initial phase of analyzing and modeling the system, the UML diagrams of the Campus Lost and Found System were reviewed and improved. Various improvements were carried out to make sure that the UML diagrams depict system behavior in an accurate manner.

Use case diagram:

- User roles divided into Finder and Owner
- <<include>> relationships added for reusable functionality
- Improved use case relationships
- Better organization and readability of functions

Class diagram:

- Inheritance relationships added (user generalized into owner, finder and admin)
- Enhanced class structure
- Improved associations and multiplicities among classes

Sequence diagram:

- Activation bars added for processes
- Message sequence improved
- Alt blocks included for: Claim approved / claim rejected
- Improved interactions between actors, system, and database

Activity diagram:

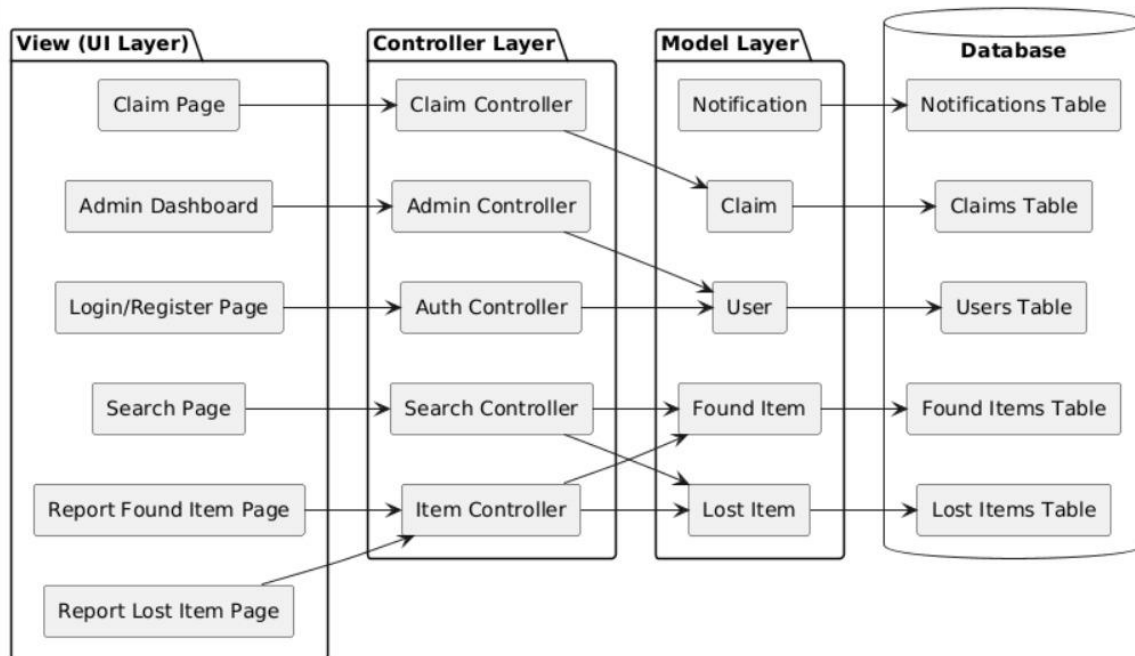
- Additional decision points were added
- Validation procedures were included
- Alternate flows were included for invalid input
- Improvements were made to process flow

State diagram:

- Clear item states were defined
- State transitions were improved
- Procedure for handling refused claims was included

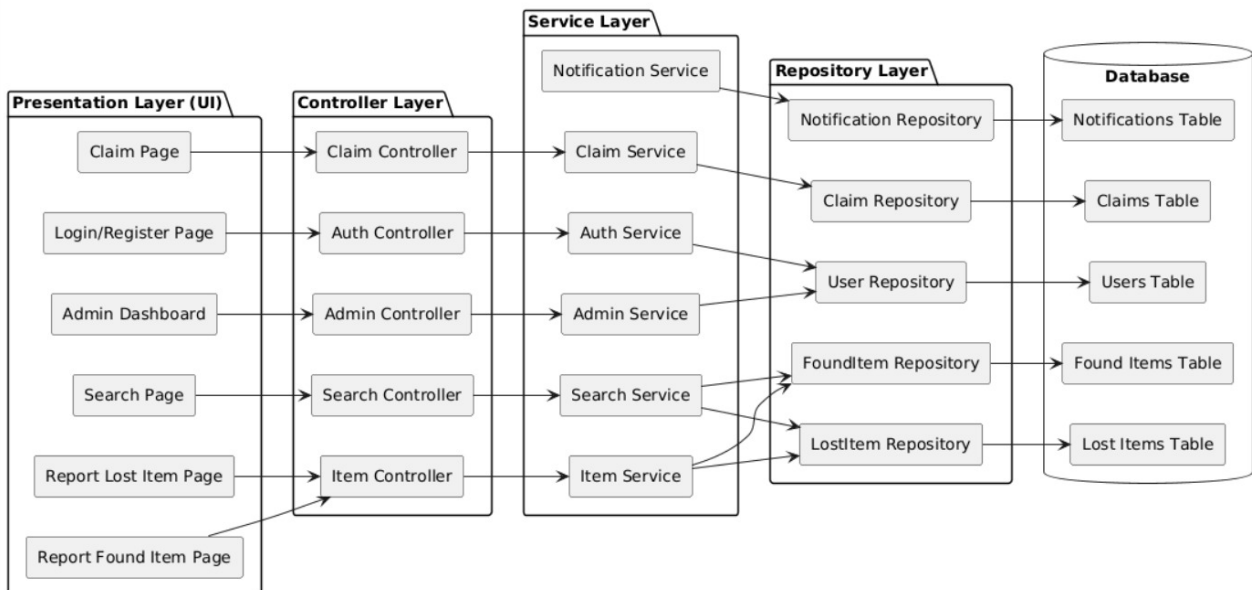
MVC Pattern:

MVC Architecture - Digital Lost & Found System



Layered Pattern:

Layered Architecture - Digital Lost & Found System



Testing:

- **Test case 1: Login**

Test Case ID	TC-01
Use Case ID	UC-01
Title	User Login
Description	Checks if a registered user can log into the system successfully.
Preconditions	User is already registered in the system.
Test Steps	1. Open login page 2. Enter valid email and password. 3. Click login button.
Expected Result	System authenticates the user and grants access successfully.
Status	Pass

- **Test case 2: Report Found Item**

Test Case ID	TC-02
Use Case ID	UC-02
Title	Report Found Item
Description	Check whether user can successfully report a found item with valid details.
Preconditions	User is logged into the system.
Test Steps	1. Opens form 2. Enters valid item details (name, description, location, date) 3. Clicks Submit
Expected Result	System saves the item successfully and shows confirmation message
Status	Pass

- **Test case 3: Search lost item**

Field	Details
Test Case ID	TC-03
Use Case ID	UC-03
Title	Search Lost Item
Description	Checks whether Owner can search for lost items.
Preconditions	Owner is logged into the system.
Test Steps	1. Enter item keyword or category. 2. Click search button.
Expected Result	System displays matching found items.
Status	Pass

- **Test case 4: Approve Claims**

Field	Details
Test Case ID	TC-4
Use Case ID	UC-8
Title	Approve Claims
Description	Test whether admin can approve submitted claims
Preconditions	Admin must be logged in and claim must exist
Test Steps	1. Opens claims dashboard 2. Selects a claim 3. Reviews details 4. Clicks approve or reject
Expected Result	Claim status is updated successfully and user is notified
Status	Pass

References

- [1] V. P. I. Ltd., "Visual Paradigm".
- [2] Rimonkumar, "Lost and Found Management System," [Online]. Available:
<https://github.com/Rimonkumar/Lost-and-Found-Management-System/blob/main/Lost%20and%20Found%20Management%20System%20.pdf>.
- [3] J. H. Jibon, "Lost and Found – Web Project," SlideShare, 2019.
- [4] C. R. C. Siok Yee Tan, "An Effective Lost and Found System in University Campus," *Journal of Information System and Technology Managemen*, 2023.
- [5] A. Roques, "PlantUML".